BITS Pilani – Work Integrated Learning Programmes
Introduction to Statistical Methods
Course No.: AIMLCZC418 — Set 1

Instructions: Answer all questions. Show all working clearly. Total: 6 questions. Use of standard normal (Z) tables is permitted where required.

Question 1 [5 Marks]

a) A telecom company is assessing the performance of its customer support center using the following customer waiting times (in minutes):

2.1, 3.5, 4.0, 5.2, 6.8, 2.9, 3.0, 4.5, 7.2, 8.0, 1.5, 2.8, 3.7, 4.9, 6.0, 7.5, 2.2, 3.3, 4.1, 5.8

(i) Calculate the five-number summary for the data. [2 Marks]

(ii) Interpret the results and discuss what they indicate about: [1 Mark]

   a) Customer service efficiency.
   b) The presence of unusually long waiting times.

b) In an e-coding competition, participant A has a 50% chance of completing the final challenge. If A fails to complete the challenge, participant B wins with probability 0.85. If A completes the challenge, participant B wins with probability 0.25. Determine the probability that participant B wins the competition. [2 Marks]

Question 2 [6 Marks]

The National Testing Agency (NTA) conducted a post-examination analysis of 600 NEET 2025 aspirants to study subject-wise performance in Physics (P), Chemistry (C), and Biology (B). The following information was obtained:

$$\begin{aligned} |P| &= 380 \quad & |P \cap C| &= 150 \\ |C| &= 260 \quad & |P \cap B| &= 250 \\ |B| &= 420 \quad & |C \cap B| &= 180 \end{aligned}$$

Assume that every student qualified in at least one subject. Answer the following:

a) Determine the number and probability of students who qualified in all three subjects. [1 Mark]

b) Determine the probability that a randomly selected student qualified in exactly two subjects. [1 Mark]

c) Determine the probability that a student qualified in Physics but not Biology. [1 Mark]

d) Determine the probability that a student qualified in only Biology. [1 Mark]

e) Determine the probability that a student qualified in Physics or Biology but not Chemistry. [1 Mark]

f) Based on your calculations, comment on the overall overlap among the three subjects and its implications for student performance patterns. [1 Mark]

Question 3 [4 Marks]

A manufacturer of AI accelerator chips reports that the probability of a chip being defective is 0.03. A quality control engineer randomly inspects 40 chips from a large shipment.

(i) Find the probability that exactly 4 chips are defective. [1 Mark]

(ii) Find the probability that between 2 and 5 chips (inclusive) are defective. [2 Marks]

(iii) The shipment is flagged for further inspection if the sample contains more than 4 defective chips. Compute the probability that the shipment is flagged. [1 Mark]

Question 4 [5 Marks]

The lifetime (in years) of a certain type of battery is modelled by the probability density function:

$$f(x) = \begin{cases} \dfrac{x}{50}, & 0 \le x \le 10 \\ 0, & \text{otherwise} \end{cases}$$

(i) Is $f(x)$ a valid probability density function?

(ii) Find the probability that a battery lasts more than 8 years.

(iii) Find the expected lifetime $E(X)$.

(iv) Find the variance of X.

(v) Interpret the results in terms of product reliability.

Question 5 [5 Marks]

A company wants to classify whether a customer will **buy a product (Yes/No)** based on two features: **Age Group** and **Income Level**. Past data is summarized below:

Age Group	**Income Level**	**Buy = Yes**	**Buy = No**
Young	Low	5	15
Young	High	20	5
Middle	Low	10	10
Middle	High	30	10
Senior	Low	3	12
Senior	High	12	8

A prospective customer belongs to the **Middle** age group and has a **High** income level. Using the Naïve Bayes assumption of conditional independence:

a) Compute the prior probabilities of the target classes. [1 Mark]

b) Calculate the likelihood probabilities associated with the customer's attributes. [1 Mark]

c) Determine the posterior probabilities for both classes (Buy = Yes and Buy = No). [2 Marks]

d) Predict whether the customer is likely to purchase the product and interpret the result from a business perspective. [1 Mark]

Question 6 [5 Marks]

a) A bank uses an automated system to flag potentially fraudulent transactions. From historical data, 2% of all transactions are fraudulent. If a transaction is fraudulent, the system correctly flags it 90% of the time. If a transaction is not fraudulent, the system incorrectly flags it as suspicious 5% of the time. One day, a customer's transaction is flagged as suspicious. [3 Marks]

(i) Compute the probability that the flagged transaction is actually fraudulent.

(ii) Comment on whether the system's flag strongly indicates fraud, and discuss implications for customer trust.

b) An e-commerce company records the delivery times (in days) for its packages. Data analysis shows that delivery times follow a normal distribution with a mean of 5 days and a standard deviation of 1.2 days. [2 Marks]

(i) What is the probability that a package is delivered in less than 4 days?

(ii) Find the probability that a package is delivered between 4 and 6 days.